PBL PROJECT · INDIAN NUMBER PLATE RECOGNITION

RISKGATE ANPR

A local number plate reader that knows when to ask for review.

Detect → Read → Estimate correctness → Accept or review

01 / PROBLEM

A plausible answer can still be wrong.

First prototype

Send a test photo to Gemini and display its answer.

Final question

Can a local system tell when its plate reading is safe enough to use?

A one-character mistake can make a different vehicle's plate. Format checks alone cannot catch it.

02 / SYSTEM

Five understandable steps.

1 Detect the plate →

2 Read the crop →

3 Check format →

4 Estimate correctness →

5 Accept / review

Everything runs locally. The demo shows the photo, box, crop, text, confidence, decision, and a correction input.

03 / DATA AND METHOD

Real images, grouped evaluation.

1,695
usable image / XML text pairs

1,027
detector training images after expansion

12
full vehicle scenes in the demo gallery

The original 254-image test stays separate from training. Repeated plates and source-video frames are grouped; a second box-only source added 180 screened full images for detector training.

Sources: Saisirishan and Kedarsai Kaggle datasets. XML labels have only a sampled audit.

04 / MODEL STACK

Small models, clear roles.

YOLO11n

Fine-tuned to find the plate rectangle in a vehicle image.

FastPlateOCR

Selected against EasyOCR on the same training crops; reads the plate locally.

Logistic model

Uses OCR and image signals to estimate whole-plate correctness.

The system does not depend on a paid API or internet connection.

05 / MEASURED RESULT

Finding works better than reading.

248 / 254

expanded detector matches at IoU ≥ 0.5; original 237

71 / 254

full-image plates read exactly

28.0%

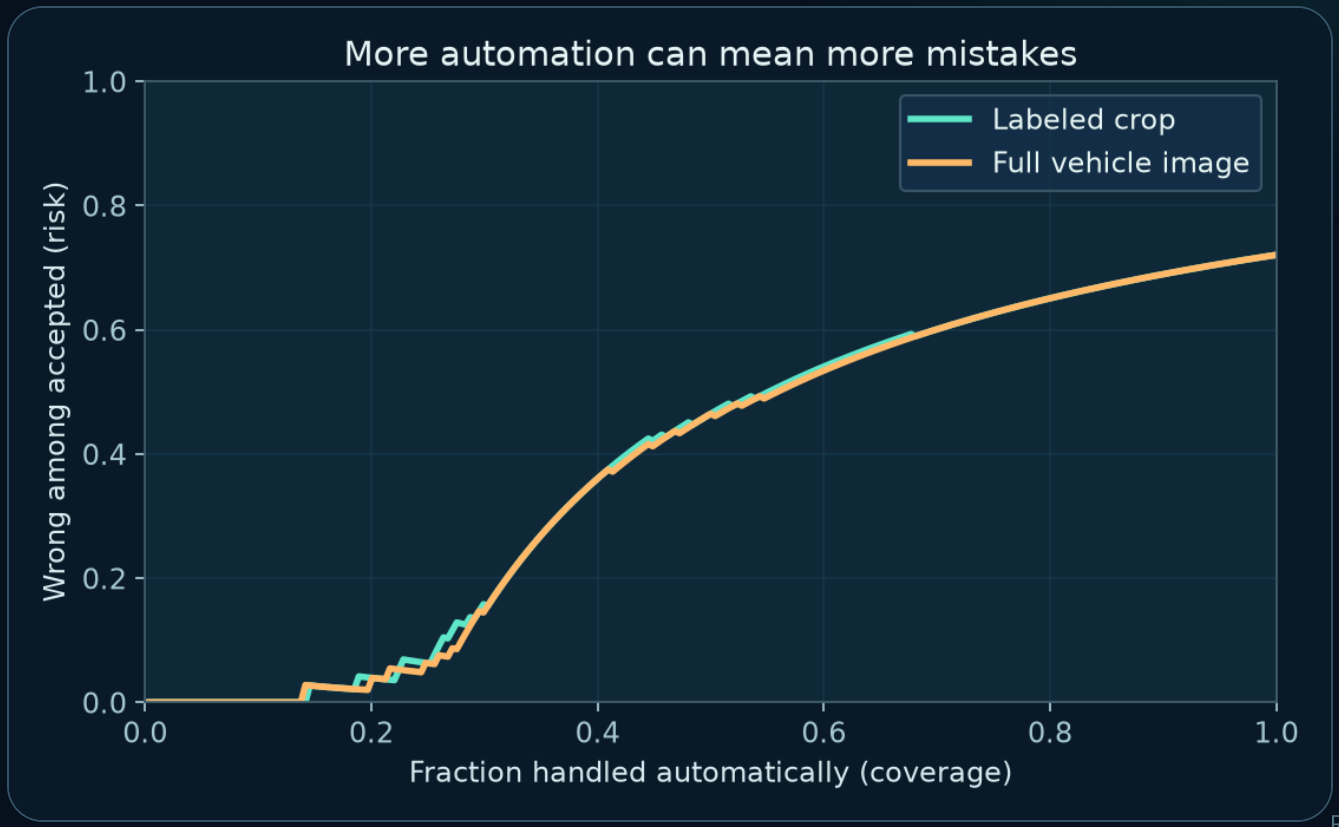
end-to-end exact match

Extra detector data improved localization on a repeated test, while exact OCR stayed the bottleneck. The second comparison is exploratory, not blinded.

Automation has a cost.

At the threshold chosen for a 10% validation target, the full pipeline accepted 75 / 254 images automatically. 11 / 75 accepted answers were wrong.

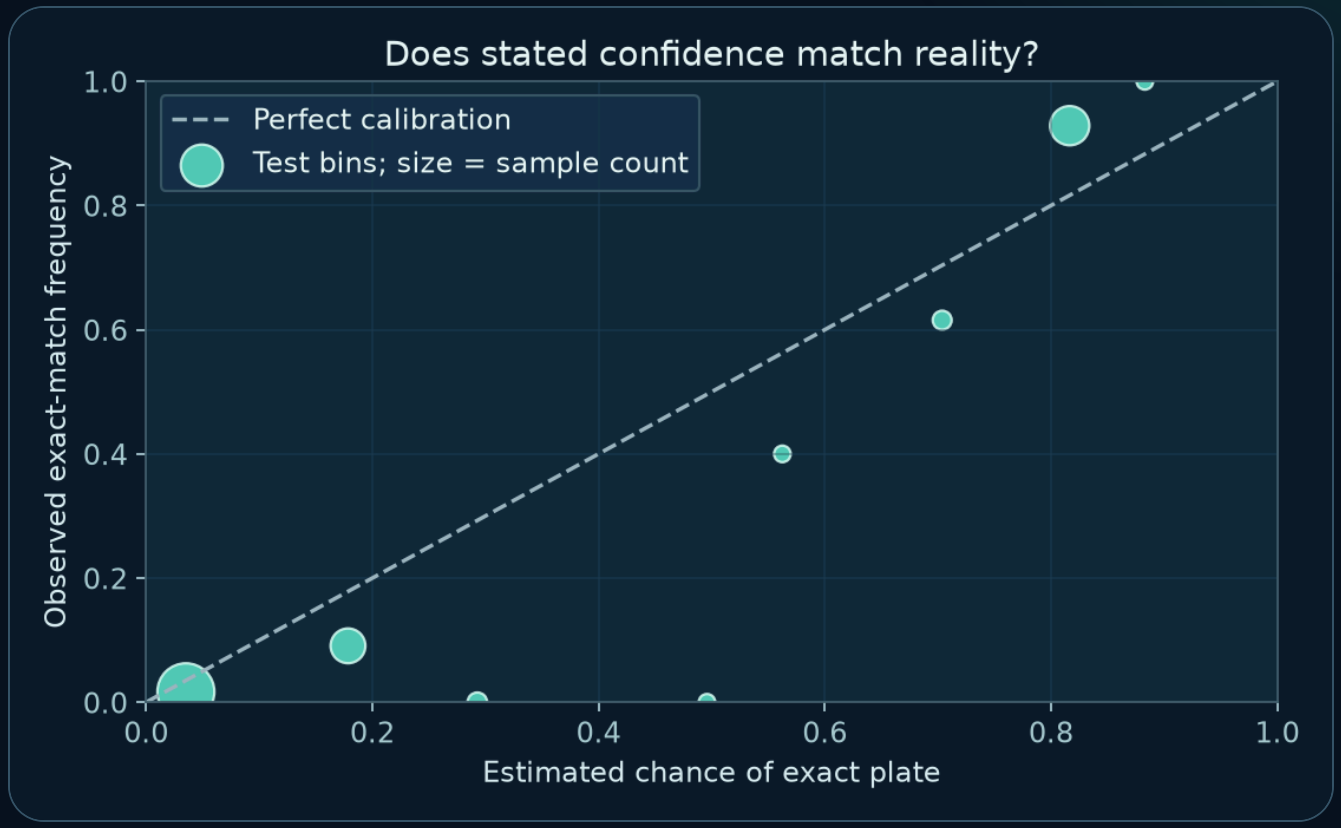
Observed accepted error: **14.7%**. The target was not met on test. Strict 2% and 5% targets accepted none.



Confidence is tested, not assumed.

A raw OCR score of 90% does not prove a 90% chance that the entire plate is right.

On the crop test, calibration ECE changed from 0.559 (raw OCR score) to 0.075 (correctness estimate). The plot checks whether confidence matches observed frequency.



08 / LIMITS AND NEXT STEP

What this prototype proves.

Done

Offline full-image pipeline, detector training, two-reader comparison, calibrated routing, human correction, real test charts.

Limits

Exact OCR is 28%. A 180-crop audit found 8 clear data issues. Some Google images show foreign plates. Human review is still necessary.

Next: curate Indian-only labels and train a stronger OCR model. Keep the same grouped evaluation to measure whether it really helps.

Research basis: BLPR, Neural Sentinel, ICPR low-resolution LPR, and the other sources in docs/EVALUATION_REPORT.md.